



Brazilian E Commerce Analytics:

A Data-Driven Growth Strategy

Made By
Justin Varghise

Olist Marketplace (2016-2018

96,000+ Orders | \$13.5M Revenue | 93K Customers)

Tools: SQL • Python • Excel • Power BI



Executive Summary

The Objective

Analyze Olist's e-commerce data to identify growth opportunities and operational inefficiencies.

Built end-to-end analytics framework using SQL, Python, Excel, and Power BI.

Key Market Insight

3% repeat rate (vs 25-30% industry standard)

85% revenue from 5 states (Southeast concentration)

12.5-day avg delivery (vs 7-10 day standard)

40% revenue in Q4 only

Recommendations

4-priority action plan targeting retention, expansion, delivery, and seasonality.

Projected Impact: \$2.27M annual benefit 425% 3-year ROI 18-month payback

Olist's Growth Problem: Strong Sales, Weak Foundation

Retention Crisis

Only 3% repeat purchases
97% one-time buyers
CLV: \$141 vs CAC: \$50

Geographic Risk

85% revenue from 5 states
North/Central-West: <5%
Over-reliance on São Paulo

Delivery Gaps

Average: 12.5 days 10%
delayed orders Delayed =
-2.3 star rating

Seasonal Volatility

40% revenue in Q4 (Nov-
Dec) 25% revenue drop in
Q1 Cash flow
unpredictability

Not All Customers Are Equal: RFM Segmentation Results

Segment	Volume	Key Behaviors	Business Implication
Champions	18% (17K)	- High frequency, recent - Avg CLV: \$850 4+ orders	These are your VIPs. Build retention programs to keep them engaged and drive referrals.
Loyal Customers	22% (20K)	- Regular buyers - Avg CLV: \$420 2-3 orders	Key retention target. Move them to Champions with rewards and personalization.
Potential Loyalists	25% (23K)	- Recent customers - Avg CLV: \$280 1-2 orders	New customers showing promise. Nurture with follow-up campaigns and discounts.
At Risk	15% (14K)	- Declining engagement - Avg CLV: \$380 - Last purchase >6 months	Win-back campaigns needed. Offer incentives to re-activate before churn.
Need Attention	20% (19K)	- One-time buyers - Avg CLV: \$140 - Low intent	Require strong value proposition to convert. Low priority for spend.

Key Takeaway: 40% of customers (Champions + Loyal) drive 65% of revenue. Retention strategy must prioritize these segments.

85% Revenue from 15% of Brazil's Geography

Top 5 States

(85% of Revenue)

1. Sao Paulo (SP): \$5.7M (42%)
2. Rio de Janeiro (RJ): \$1.8M (13%)
3. Minas Gerais (MG): \$1.6M (12%)
4. Rio Grande do Sul (RS): \$950K (7%)
5. Paraná (PR): \$810K (6%)

Underserved Regions

North Region:

- 2% revenue
- 18M population
- 20-25 day delivery

Central-West Region:

- 3% revenue
- 16M population
- 15-18 day delivery

Opportunity

- Current Coverage: 60% of Brazil's population
- Target Coverage: 95%
- Addressable Market: +40M potential customers
- Projected Revenue: +\$4M annually

What's Needed: Regional fulfillment centers + local logistics partnerships

85% Revenue from 15% of Brazil's Geography



On Time Delivery

- 4.7 star avg rating
- 82% positive reviews
- High satisfaction

Delayed Delivery

- 2.4 star avg rating
- 64% negative reviews
- High complaints

Current State

- Average Delivery: 12.5 days
- Industry Standard: 7-10 days
- On-Time Rate: 90%
- Delayed Orders: 10% (9,600 orders)

Regional Disparities:

- Southeast (SP/RJ/MG): 8-10 days
- South (RS/SC/PR): 10-12 days
- Northeast (BA/PE): 15-18 days
- North (AM/PA): 20-25 days

The Fix

- Target: 12.5 → 8 days average
- How: AI routing + Same-day delivery in metros + Delivery guarantees
- Result: 90% → 95% on-time | 4.1 → 4.5 stars | 3% → 6% repeat rate



Top Categories Drive 60% of Revenue

Category Performance

High Performers (Growing)

- Electronics: +50% YoY growth
- Health & Beauty: Highest volume
- Sports: Strong margins

Underperformers(Declining)

- Fashion: High return rate
- Books: Low margins
- Food: Logistics challenges

Product Mix Strategy

Concentration risk

- Top 3 categories = 24% revenue
- Over-reliance on Health/Beauty/Home
- Diversification needed

Opportunity

- Electronics growing fastest
- Cross-sell potential (Home + Garden)
- Bundle opportunities

Top 3 Best-Selling Products

Wellness/Beauty Kit

- Category: Health & Beauty
- Revenue: \$180K | Orders: 4,200
- AOV: \$42.86 | Rating: 4.6

Bed linin set(Queen)

- Category: Bed/Bath/Table
- Revenue: \$165K | Orders: 2,800
- AOV: \$58.93 | Rating: 4.7

Fitness Tracker Watch

- Category: Sports & Leisure
- Revenue: \$145K | Orders: 1,850
- AOV: \$78.38 | Rating: 4.5

RECOMMENDATION: Double down on Top 3 categories, Expand Electronics,. Reduce low-margin categories, Test product bundles

Strength

Strong Southeast Market Presence

- 85% market share
- Established brand recognition

Diverse Product Catalog

- 70+ categories
- Health/Beauty, Home top performers

High Customer Satisfaction

- 4.7 stars for on-time delivery
- 75% rate 4-5 stars overall

Flexible Payment Options

- Credit card installments (3.5 avg)
- Boleto for unbanked customers

Weakness

Customer Retention Crisis

- 3% repeat rate vs 25-30% standard
- No loyalty program or engagement

Delivery Performance Gaps

- 12.5 days avg vs 7-10 day standard
- 10% delayed orders

Geographic Concentration Risk

- 85% revenue from 5 states only
- Over-reliant on São Paulo (42%)

Seasonal Revenue Dependency

- 40% revenue in Q4 (Nov-Dec)
- 25% Q1 revenue drop, cash flow issues

Opportunities

Untapped Geographic Markets

- North/Central-West: <5% penetration
- 30%+ of population underserved
- Potential: +\$4M annual revenue

Customer Retention Programs

- Loyalty rewards → 15% repeat rate
- Personalization + subscriptions
- CLV improvement: \$141 → \$350

Same-Day/Next-Day Deliver

- Major metros ready for fast shipping demand
- Competitive differentiation
- Premium pricing tier opportunity

Category Expansion & Data Insights

- Electronics growing 50% YoY
- B2B wholesale opportunities
- Data monetization to sellers

Threats

Intense Competition

- Amazon entering Brazil
- Shopee's aggressive pricing

Economic Volatility

- Brazil GDP fluctuations
- Currency devaluation
- Inflation reduces purchasing power

Rising Customer Expectations

- "Amazon effect" - faster delivery
- Free shipping expectations
- Higher service standards required

Logistics Infrastructure Challenges

- Poor roads in remote areas
- Strike risks (truckers, postal)
- Rising fuel costs

Strategic Roadmap

PRIORITY 1: Customer Loyalty Program

Investment: \$500K | Timeline: 6 months

Actions:

- Tiered rewards (Bronze/Silver/Gold)
- Personalized email campaigns
- Subscription service for recurring products

Impact: Repeat rate 3% → 15% | CLV \$141 → \$350 | +\$1.5M revenue (estimated)

PRIORITY 3: Delivery Excellence

Investment: \$750K | Timeline: 9 months

Actions:

- AI-powered routing optimization
- Delivery time guarantees with compensation
- Same-day/next-day in major metros

Impact: Avg delivery 12.5 → 8 days | On-time 90% → 95%(estimated)

PRIORITY 2: Geographic Expansion

Investment: \$2M | Timeline: 12 months

Actions:

- 3 new fulfillment centers
- Regional logistics partnerships
- Localized marketing campaigns

Impact: Coverage 60% → 95% | +\$4M revenue | -40% delivery time (new regions) (estimated)

PRIORITY 4: Revenue Stabilization

Investment: \$200K | Timeline: Ongoing

Actions:

- Off-season promotional campaigns
- Event-based marketing
- B2B wholesale program

Impact: Q1 dip 25% → 10% | Monthly variance -40% | Cash flow +60% (estimated)

**Thank
You!**